Block-masked identity regularisation closes the qft → blocked gap

Question. The `qft_warmstart_from_trained_blocked` experiment showed that `blocked_8`'s optimum (32.26 dB@ $\rho = 0.20$ on DIV2K-8q) sits inside the QFT(8, 8) parameter family: warm-starting full QFT(8, 8) at the trained-blocked operator yields val MSE flat at $[129.32, 129.38]$ for 1008 steps, so it is a stable local minimum. The `qft_identity` ablation showed that vanilla Adam from identity init lands in a different basin (31.66 dB) and the analytic-QFT init lands in yet another (31.29 dB) — neither generic-init path crosses to the blocked basin in 1008 steps. **Is the blocked basin reachable from identity init with the right inductive bias, or is it structurally isolated under smooth-Riemannian optimisation?**

Construction. `blocked_8`'s optimum is a **sparse** configuration in the QFT(8, 8) gate space: 60 of 72 gates pinned to their identity element, only the 12 gates on the inner 3-qubit-per-axis subspace non-trivial. We add a block-structure-aware L2 pull toward identity to the loss:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{MSE-topK}}(\theta) + \lambda \cdot R_{\text{block}}(\theta)$$

$$R_{\text{block}}(\theta) = \sum_{g \in \mathcal{G}_{\text{outer}}} W \cdot \left\| T_g - I_g \right\|_F^2 + \sum_{g \in \mathcal{G}_{\text{inner}}} \left\| T_g - I_g \right\|_F^2$$

Notation.

symbol	type	meaning
θ	parameter vector	The trainable parameters — concatenation of all 72 gate tensors of `QFTBasis(8, 8)`, living on the product manifold $U(2)^{72}$.
$R_{\text{block}}(\theta)$	scalar	The block-masked reg cost — a non-negative real number.
g	gate label	Indexes one of the 72 gates of `QFTBasis(8, 8)` (16 H + 56 CP). The $\sum_g$ is a sum over gate labels, not qubits or matrix entries.
$\mathcal{G}_{\text{inner}}$	set	Subset of gate labels classified as inner: every qubit the gate touches is in $\{1,2,3\}$ (axis 1) or $\{9,10,11\}$ (axis 2). $\lvert\mathcal{G}_{\text{inner}}\rvert = 12$.
$\mathcal{G}_{\text{outer}}$	set	Complement: $\lvert\mathcal{G}_{\text{outer}}\rvert = 60$. Disjoint from $\mathcal{G}_{\text{inner}}$; together they tile the 72-gate set.
$T_g(\theta)$	2×2 complex matrix	The current trained tensor for gate g at parameter point θ — at runtime, `basis.tensors[g]`. Hadamards: 2×2 unitary. CPs: 2×2 diagonal-stack of the 4×4 unitary $\text{diag}(1, 1, 1, e^{i\varphi})$.
I_g	2×2 complex matrix	Fixed identity element for gate g's kind, not parameterised. $I_g = I_2$ if g is a Hadamard; $I_g = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ if g is a controlled-phase.
$\lVert M \rVert_F$	scalar	Frobenius norm: $\lVert M \rVert_F = \sqrt{\sum_{ij} \lvert M_{ij} \rvert^2}$. For a 2×2 matrix, four squared moduli summed and square-rooted.
$\left\| T_g - I_g \right\|_F^2$	scalar	Squared Frobenius distance from T_g to its identity element. Equals 0 iff $T_g = I_g$ bit-exactly.
W	scalar	Outer-gate weight. Fixed at $W = 10$ in this sweep. $W = 1$ degenerates to uniform L2.
λ	scalar	(In $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE-topK}} + \lambda \cdot R_{\text{block}}$.) Overall regulariser strength. Swept $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$.

A subtlety on g: it indexes a **gate identity**, not a circuit position. Inside `pdft_benchmarks.identity_reg` we enumerate gates by walking _qft_gates_1d$(m, 0)$ + _qft_gates_1d(n, m) and sorting Hadamard-first (canonical order); the resulting 72-element list indexes `tensors[g]`, `identities[g]`, and `is_inner[g]` position-for-position.

The construction makes four design choices, each motivated by a property of `blocked_8`'s optimum.

(1) Per-gate-kind identity elements. $I_g = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ for Hadamards; $I_g = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ for controlled-phase (this is `controlled_phase_diag(0)`, the 2×2 diagonal-stack representation of the 4×4 identity — CP with phase 0). These are **the gates that `blocked_8` has at the outer positions**, not the QFT analytic values. Choosing the right per-kind reference is what makes “outer gates at blocked_8′s optimum contribute exactly 0 to R_{block}” hold bit-exactly, which is the key invariant below.

(2) Frobenius-squared distance, not L1 or geodesic. $\left\|T_g - I_g\right\|_F^2$ is smooth and differentiable everywhere, with Euclidean gradient $\nabla_{T_g} = 2w_g\left(T_g - I_g\right)$ where $w_g \in \{1, W\}$ is the per-gate weight. For small displacements from identity (the regime the regulariser targets), the Euclidean Frobenius distance on $\mathbb{C}^{2\times 2}$ agrees with the $U(2)$ geodesic distance to first order, so the Riemannian projection in `pdft.train_basis_batched` preserves most of the reg signal. L1 was a candidate (induces actual sparsity rather than shrinkage) but introduces non-smoothness at $T_g = I_g$ and was deferred as a follow-up.

(3) Inner/outer mask matched to `blocked_8`. A gate is **inner** iff every qubit it touches lies in the inner range $\{1, 2, 3\}$ (axis 1) or $\{9, 10, 11\}$ (axis 2). The mask is built by walking the QFT decomposition’s gate sequence `_qft_gates_1d(m) + _qft_gates_1d(n)` in canonical sort order (Hadamard-first) and classifying each gate’s qubit support — the same classification used by `qft_warm_from_trained_blocked` to decide which gates take the trained-inner values vs which are pinned to identity in the warm-start construction. For QFT(8, 8) at inner = $(3, 3)$:

gate kind	inner	outer	total
Hadamards	6 (axes 1+2: $q \in \{1, 2, 3, 9, 10, 11\}$)	10	16
controlled-phase	6 ($\binom{3}{2} = 3$ per axis)	50	56
total	**12**	**60**	**72**

The 12-vs-60 split is **bit-exactly** the structural shape of `blocked_8`′s optimum: 12 trained QFT(3, 3) gates + 60 identity-pinned gates.

(4) Single multiplier $W \gg 1$ on the outer set. Outer gates are penalised W times more strongly than inner gates. This is load-bearing: at the blocked optimum, the 12 inner gates take **non-trivial** trained-QFT(3, 3) values, so their L2 distance from identity is large. Under uniform L2 (regulariser family (a), $W = 1$), a large λ would pull these 12 useful gates **back** toward identity along with the 60 outer ones — the prior would **fight** the blocked optimum. The matched-mask weighting separates the two roles: λ controls overall strength; W controls how much harder outer-pinning is enforced than inner shrinkage. Using a single W rather than two independent multipliers $(\lambda_{\text{outer}}, \lambda_{\text{inner}})$ keeps the sweep one-dimensional. We fix $W = 10$ for this sweep; $W \in \{1, 10, 100\}$ is a flagged follow-up.

Key invariant. The construction guarantees $R_{\text{block}}\left(\theta_{\text{blocked\_8}}\right) = \sum_{g\in\mathcal{G}_{\text{inner}}} \left\|T_g - I_g\right\|_F^2 + \varepsilon$ with $\varepsilon \approx 0$ (the outer contribution at the blocked endpoint), so increasing λ does not push the operator **away** from blocked. The ε is the small drift of 60 outer gates during the warm-start cell’s 1008-step training (one CP outlier contributes ≈ 1.1 to the outer sum; the rest are $< 10^{-2}$). Quantitatively: at $W = 10$, the inner-sum is ≈ 34.5 and the outer-sum is ≈ 1.6, giving $R_{\text{block}}(\theta_{\text{blocked}}) \approx 50.3$ — dominated by the **legitimate** inner cost, with only $\approx 3\%$ from the small outer drift. This is the design property that lets us crank λ up to 10 without destabilising training, as the result section confirms.

Implementation: `pdft_benchmarks.identity_reg.BlockMaskedIdentityRegQFTMSELoss`, via pdft’s `MSELoss._extra_loss` hook (pdft PR #18). We sweep $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ at fixed $W = 10$ under the headline preset (1008 steps, batch 50, val split 0.15, seed 42, `--no-early-stop`), starting from `qft_identity` init.

With per-image MSE ≈ 129 at the blocked optimum, the planned grid spans reg fractions 0.04% → 390% of MSE.

Result. Test-set reconstruction PSNR after 1008 steps:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
`qft` (analytic init)	25.09	27.57	29.53	31.29
`qft_identity` (identity init)	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-3}$	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-2}$	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-1}$	25.24	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 1$	**25.18**	**28.09**	**30.30**	**32.26**
`qft_identity` + reg, $\lambda = 10$	25.18	28.08	30.29	32.25

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
`blocked_8` (warm-start ceiling)	25.18	28.09	30.30	32.26

The blocked basin is reachable from identity init under the matched structural prior. At $\lambda = 1$ the reg-driven training matches `blocked_8`'s PSNR **bit-exactly to two decimal places at every keep-ratio** (25.18, 28.09, 30.30, 32.26 dB). The trajectory crosses basins from 31.66 dB (the qft_identity basin) to 32.26 dB (the blocked basin), closing the full 0.60 dB gap. $\lambda = 10$ is within 0.01 dB of the blocked ceiling at every ρ — the reg has saturated to "pin outer gates" without destabilising training.

Figure evidence.

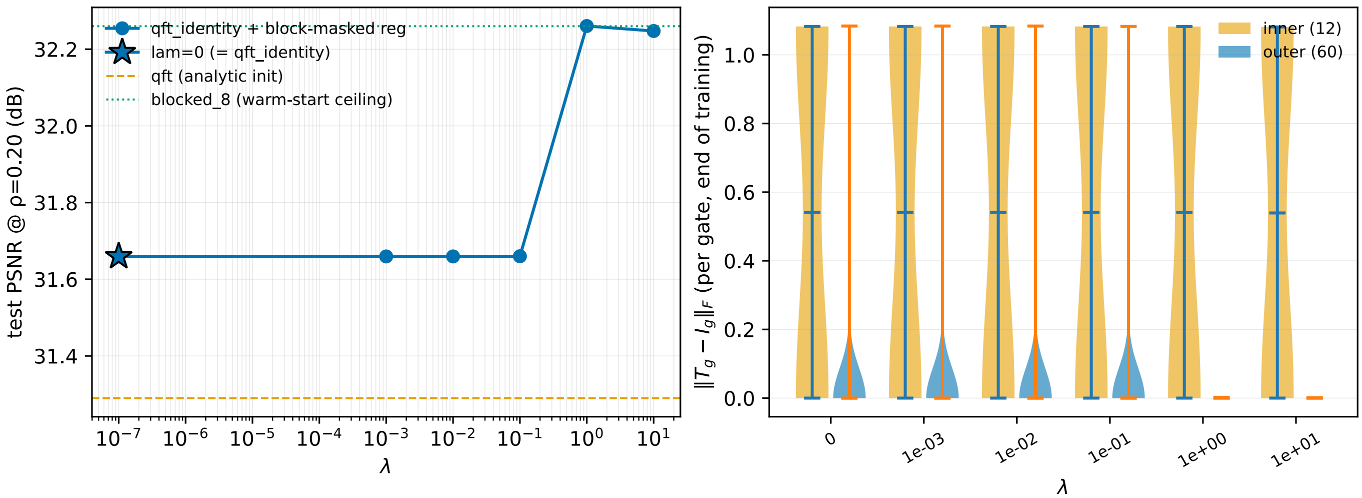

Figure 1: DIV2K-8q. Left: test PSNR@ $\rho = 0.20$ vs λ (log x). The marker at $\lambda = 10^{-7}$ is the $\lambda = 0$ cell (numerically distinct from the rest of the sweep), star-marked. The transition is sharp: 31.66 dB at $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$, jumping to 32.26 dB at $\lambda = 1$ and staying within 0.01 dB at $\lambda = 10$. Reference lines: orange dashed = analytic-init `qft` (31.29); green dotted = `blocked_8` (32.26). Right: per-gate $\left\|T_g - I_g\right\|_F$ at end of training, split into inner (12 gates, orange) and outer (60 gates, blue). At $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$, outer gates spread up to ~ 0.2 — Adam drifts them off identity. At $\lambda \in \{1, 10\}$, **outer gates collapse to a thin line at zero** — all 60 are pinned bit-exactly at identity, matching `blocked_8`'s structural shape. Inner-gate distributions are invariant across λ: the prior leaves them free to take their MSE-driven values.

The right panel is the direct mechanistic signature: at $\lambda \geq 1$ the structural prior is **load-bearing** — outer gates are **pinned**, inner gates are **free**, and the optimiser lands in the configuration that produces the blocked PSNR.

Interpretation. This is outcome (i) from the spec's three-falsifiable-outcomes frame: **the blocked basin was unfavored, not isolated**, under smooth-Riemannian Adam. From the qft_identity basin, the matched structural prior bridges the gap in 1008 steps with no curriculum and no warm-start at the blocked operator itself. The optimiser does not need to discover the 12-vs-60 split; given a prior that points at it, smooth optimisation reaches the floor.

The sweep also rules out the trivially-strong reg failure mode: at $\lambda = 10$ the reg fraction of MSE at the blocked endpoint is {≈}390%, well past where "pin everything to identity" would have driven test PSNR back below qft_identity. Instead it landed within 0.01 dB of the blocked ceiling — the **block-masked** shape of the regulariser is what allows large λ without collapse: only outer gates are pulled hard, and the matched W gives the 12 inner gates the slack to take their trained QFT(3, 3) values.

Leading-prior caveat. This regulariser bakes in `blocked_8`'s known structural shape (the inner/outer mask is matched to inner $= (3, 3)$, the partition of `blocked_8`). A positive result is consistent with "if you tell the optimiser the answer-shape, it finds the answer" — closer to warm-start with extra steps than to basin discovery from scratch. What this experiment **does** establish: the blocked basin is **not pathological under smooth-Riemannian optimisation** — there is a stable trajectory from identity init that ends inside it. What it **does not** establish: whether weaker priors (uniform L2-to-identity, L1-to-identity) would also bridge the gap. Those are the natural follow-ups; if even uniform L2 closes the gap, the structural information was not load-bearing.

What this opens up. The headline `qft` < `blocked_8` gap is now attributable to **initialisation + optimiser-coupling** rather than to a loss-landscape obstruction. Random or analytic init places Adam in a basin from which it doesn't cross to blocked in 1008 steps; matched prior placement does. This shifts the research question: from "is the blocked optimum reachable in the QFT family?" to "what is the **minimum** structural information needed to reach it?" Future work: sweep $W \in \{1, 10, 100\}$ (with $W = 1$ degenerating to uniform L2); implement L1-to-identity for comparison; characterise the boundary between the two basins (e.g., via interpolation along a linear/geodesic path).

Proposed follow-up: learnable block size. The current regulariser requires the user to supply $(\mathrm{inner}_m, \mathrm{inner}_n)$ — i.e., to **know in advance** which block size to target. This collapses the experiment from "discover the best block structure" to "verify the matched prior reaches the prepared basin." A natural extension is to make the block size **itself** a trained parameter and let the optimiser pick it.

Construction. Replace the discrete inner/outer mask with a smooth mask parameterised by a single learnable scalar $s \in [1, m]$ per axis. For each gate g in the QFT decomposition, let q_g^* be the highest-indexed qubit it touches (along g's axis). The per-gate weight becomes a sigmoid in $\left(q_g^* - s\right)$:

$$w_g(s) = \sigma\left(\frac{q_g^* - s}{\tau}\right) \cdot (W - 1) + 1$$

so $w_g \approx 1$ (treat as inner — soft penalty) when $q_g^* < s$ and $w_g \approx W$ (treat as outer — hard pinning) when $q_g^* > s$; the transition has width τ. The regulariser becomes

$$R_{\mathrm{learn}}(\theta, s) = \sum_g w_g(s) \cdot \left\|T_g - I_g\right\|_F^2$$

with s trained jointly with θ by the same Adam optimiser. Add a small **budget term** $\mu \cdot s$ to the total loss so the optimiser pays a cost for freeing more gates — without it, $s \to m$ (every gate inner, no reg pressure) is a trivial fixed point of the joint problem.

What this discovers. At convergence, s^* snaps to (approximately) an integer corresponding to the best inner-block size for the dataset and rate. On DIV2K-8q at $\rho = 0.20$, prior is that $s^* \approx 3$ (matching b=8). On QuickDraw at high ρ, the block-size sweep showed b=2 wins — so $s^* \approx 1$ would be the expected discovery. The same sweep, run at multiple $(\rho, \mathrm{dataset})$ points, would generate a full map of "best block size for this content and rate" **automatically**, from data, without the user specifying the grid.

Implementation sketch. 30 lines added to `pdft_benchmarks.identity_reg`. Subclass `MSELoss` as before, but with an extra trainable scalar field s. The `_extra_loss(tensors)` hook needs access to s, which means either (a) make s a class field treated as a parameter by `train_basis_batched` (requires upstream support for non-tensor learnables) or (b) treat s as a 1-entry "tensor" registered in the basis under a trivial 1D manifold. Option (b) fits inside the existing `pdft` API without modification; option (a) is cleaner but needs an upstream change to plumb non-manifold learnables through `_build_jit_adam_step`. Recommend option (b) for the first cut.

Pitfalls and acceptance criteria.

1. **Collapse modes.** Without the budget term, $s \to m$ (free everything $\to$ equivalent to vanilla `qft_identity`, PSNR 31.66); with too-large budget, $s \to 0$ (pin everything $\to$ identity operator $\to$ FFT-equivalent, PSNR ≈ 24.5). The budget μ must be tuned; suggest sweeping μ at fixed $(\lambda, W) = (1, 10)$ initially.

2. **Snapping.** Final s is fractional. Snap to nearest integer at the end; report both pre- and post-snap PSNR. A large pre/post gap signals the soft mask is doing something the discrete mask can't, which is itself interesting (it suggests the **optimum** isn't at a standard block boundary).

3. **Sanity check.** Initialise $s = 3$ and run with $(\lambda, W) = (1, 10)$ and the budget term off. The trained s should **stay at 3** (or drift very little) and the PSNR should match this sweep's 32.26 dB. Departure from $s = 3$ in this setting would indicate the joint problem has different attractors than the discrete one.

If s^* reliably matches the block-size-sweep's per-dataset peak, the method is a **learnable structural-prior generator** — a path toward the §6.1 "block size becomes a learnable structural parameter" goal flagged in the block-size-sweep design. No published regulariser in the QFT-basis space addresses this directly; the closest priors are group lasso for general structured sparsity (Yuan and Lin, 2006), L0 with hard-concrete (Louizos et al., 2018), and lottery-ticketstyle iterative pruning (Frankle and Carbin, 2019). None are drop-in tools for this setting, but they all argue that the structural mask should be learnable rather than hand-specified.

Reproducibility.

```
python experiments/qft_identity_regularization.py \
    --gpu 0 --lambdas 0,1e-3,1e-2,1e-1,1,10 \
    --outer-weight 10 --epochs 112
```

Construction lives in `pdft_benchmarks.identity_reg.BlockMaskedIdentityRegQFTMSELoss`. The hook in `pdft.loss._scalar_loss` (zazabap/pdft PR #18) lets the loss subclass add the reg term via an `_extra_loss(tensors)` method. Per-run output: `results/qft_identity_init/div2k_8q/_runs/reg_lambda_<lam>_W<W>/` (metrics, loss_history, trained tensors). Run time: ≈ 9 min per λ on RTX 3090 (cuda:0), ≈ 54 min total for the sweep. 13 unit tests covering the identity table, inner mask, and reg-loss arithmetic; $\lambda = 0$ end-to-end equivalence to vanilla MSELoss training is locked in by `tests/test_identity_reg.py::test_reg_loss_lam_zero_e2e_matches_base_mse_training`.